Kedong Xiu

(修可栋)

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Education Background

Nankai University, China 2021.09-2024.06

- **Master's Degree in College of Cyber Science**
  - **Advisor:** Prof. Ding Wang
  - **Research lab:** Laboratory of Cryptography and Systems Security

Jilin University, China 2017.09-2021.06

- **Bachelor's Degree in College of Software**
  - **Major:** Pilot Program in Engineering (Software Engineering)
  - **GPA:** 3.6/4.0

Research Interests

Natural Language Processing (NLP), Human Privacy and Security, Machine Learning

Research Experience

[1] Ding Wang*, Yunkai Zou, Zijian Zhang, **Kedong Xiu**, "Password Guessing Using Random Forest," in Proc. **USENIX Security'23 (**Accepted**)**

This work opens up a new research path for constructing password guessing models by leveraging classical machine learning algorithms (e.g., Random Forest) to address the issues of data sparsity, overfitting, and low efficiency in traditional statistical and deep learning-based password guessing algorithms.

In this work, we take the *random forest* algorithm as an example to demonstrate how to transform the process of password guessing into a multiclass classification problem through re-encoding password characters, resulting in the design of a novel password guessing framework called **RFGuess**. This framework covers both trawling guessing scenarios and targeted guessing scenarios. Moreover, the new password guessing framework we proposed in this work can be expanded to a range of classical machine learning algorithms.

Status: This work has been published to **USENIX SEC'23** and got accepted.

[2] **Kedong Xiu**, Ding Wang*, "PointerGuess: Targeted Password Guessing Model using Pointer Mechanism," in Proc. **USENIX Security'24 (**Accepted**)**

In this work, we focus on the prevalent issue of password reuse and propose a novel targeted password guessing model based on users' old passwords. A user may reuse her old password through *"copy"* or *"tweak"* from the old password to create a new password. To better capture users' vulnerable reuse behaviors, we design a new targeted password guessing model using pointer networks, **surpassing state-of-the-art** in accuracy and performance.

Additionally, considering that users may *reuse multiple old passwords* and large-scale data sets are leaked, we, *for the first time*, design a new targeted password guessing model to evaluate the security of users reusing multiple old passwords, which achieves **higher** cracking rates on the same attacking scenarios.

Furthermore, we introduce a targeted password strength meter (PSM) framework that considers users' old passwords, enhancing password security evaluations. Our contributions advance password security research

significantly.
Status: We have submitted it to **USENIX SEC'24** and got accepted.

[3] Guanling Li, Ding Wang*, **Kedong Xiu**, "taPAKE: Practical Typo-Tolerant Asymmetric Password Authenticated Key Exchange," in Proc. **USENIX Security'24 (**On submission**)**

In this work, we design a typo-tolerant Password Authenticated Key Exchange (PAKE) protocol, called **taPAKE**, which enables secure communication using low-entropy passwords, even with minor spelling errors.

First, our work evaluates the accuracy of different similarity metrics in detecting users' typos. Then, we design our protocol, which utilizes cosine similarity to identify password errors while ensuring server security accurately. We provide a formal security proof within the random oracle model and demonstrate its efficiency with an execution time of only 329 milliseconds. Through extensive evaluations of a dataset of 147 million real user passwords, we validate the effectiveness and resilience of our protocol, including in targeted attack scenarios. This research significantly contributes to advancing typo-tolerant PAKE protocols, enhancing secure communication capabilities.
Status: This work has been submitted to **USENIX SEC'24.**

Publications & Under Review **(* refers to the corresponding author)**

[1] Ding Wang*, Yunkai Zou, Zijian Zhang, **Kedong Xiu**, "Password Guessing Using Random Forest," in Proc. **USENIX Security'23** (Accepted).

[2] **Kedong Xiu**, Ding Wang*, "PointerGuess: Targeted Password Guessing Model using Pointer Mechanism," in Proc. **USENIX Security'24** (Accepted).

[3] Guanling Li, Ding Wang*, **Kedong Xiu**, "taPAKE: Practical Typo-Tolerant Asymmetric Password Authenticated Key Exchange," in Proc. **USENIX Security'24** (On submission).

Rewards & Honors

Jilin University scholarship (third prize)	2017-2018
Jilin Province Mathematical Modeling Contest (second prize)	2018
Jilin University Mathematical Modeling Contest (first prize)	2018
Jilin Province Mathematical Modeling Contest (first prize)	2019
Jilin University Outstanding Undergraduate Thesis Award	2021
Nankai University Scholarship of Public Interests and All-Round Capability (third prize)	2021, 2022
"Qiangwang" Cup Artificial Intelligence Challenge Contest (second prize)	2021
"Zongheng" Cup Network Security Technology Innovation Contest (third prize)	2021
Nankai University Scholarship of Academic Progress	2022
Nankai University Scholarship of Public Interests and All-Round Capability (second prize)	2023

Personal Interests

Some sports (e.g., basketball) and Esports.
Reading: Science fiction.
Others: Working out, mountain climbing, etc.